Chess Study Assistant (CSA)

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Version 3, 29 June 2007
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manual version 1.02 Chess Study Assistant version 1.0.0

RTFM Read the fine manual (or whatever)

Why a new Chess Analysis script when there are plenty?

I hope you will feel this is a different and useful way to analyze chess PGN files. This allows you to analyze the whole game at once to a depth that you want. Analysis is saved so that it can be examined again without having to re analyze. It shows info while you are analyzing so that you can follow the process (and know it is working). Once the analysis is complete it is saved (in the saved directory) so that you will not need to analyze again. The analysis window opens letting you see some statistics about points lost and how close to Stockfish your choices were. There is a blunder count with three levels of blunder severity. All of the moves are visible and can be selected to show the game continuation. Blunders are colored according to severity in the list.

I like to have a way to display the actual continuation moves rather than just a list of moves. In the continuation screen you can try other moves with the mouse or have Stockfish calculate a new continuation, or both. None of this experimentation over writes the analysis. Continuation window can be restarted with “b” begin key to remove new moves.

Installation

Linux only! I have not made a Windows version.

Download the repository as a zip (tar) file. Unzip in the directory of your choice.
After unzipping go to the libs/stockfish_17.1 directory. Make sure the Stockfish executable has run permissions by right clicking, properties, permissions and check the allow this to run as a program. The Stockfish files included are for Ubuntu. One for older cpu’s and one for newer cpu’s. For non Ubuntu Linux systems you will have to either download the appropriate Stockfish for your Linux or set address in script to your systems Stockfish executable.

The location in the scripts for the Stockfish executable file is:
Open the chessclasses.py file in the root directory.
Go to class chess_engine()
in it’s init section is a variable stockfish_location =
Change the right side of the = to the full path of your stockfish executable.

That variable is currently set to the current working directory + "/libs/stockfish17_1/stockfish17.1-popcnt"

This defaults to the older CPU version. If you have the new CPU then change to that file.
stockfish-ubuntu-x86-64-avx2

To start script

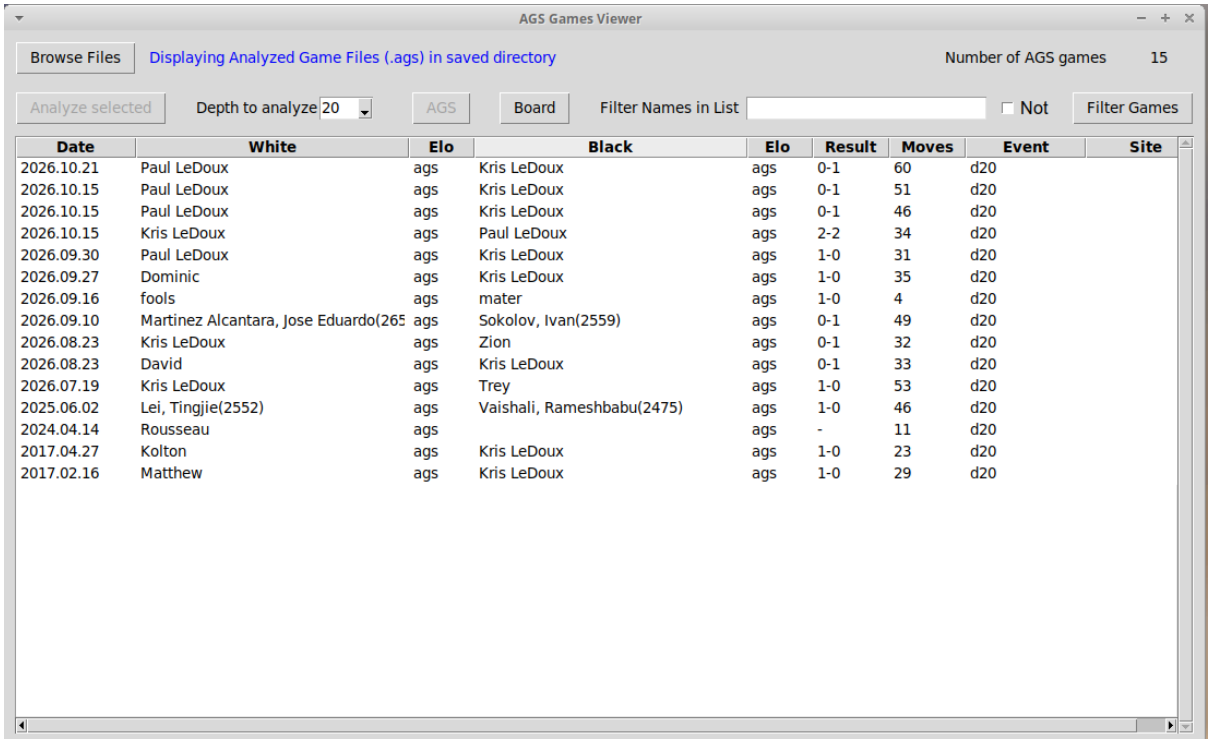
* open a terminal in the root directory that you placed (unzipped) the project.
* python3 csa.py

The non standard libraries that are not default in Python are in the libs directory included. If you already have the libraries installed and it causes interference you can start with the following command.

PYTHONPATH=. python3 -S csa.py

Operation

The script opens with the games viewer. Opens in AGS mode (analyzed game save).



Date	White	Elo	Black	Elo	Result	Moves	Event	Site
2026.10.21	Paul LeDoux	ags	Kris LeDoux	ags	0-1	60	d20	
2026.10.15	Paul LeDoux	ags	Kris LeDoux	ags	0-1	51	d20	
2026.10.15	Paul LeDoux	ags	Kris LeDoux	ags	0-1	46	d20	
2026.10.15	Kris LeDoux	ags	Paul LeDoux	ags	2-2	34	d20	
2026.09.30	Paul LeDoux	ags	Kris LeDoux	ags	1-0	31	d20	
2026.09.27	Dominic	ags	Kris LeDoux	ags	1-0	35	d20	
2026.09.16	fools	ags	mater	ags	1-0	4	d20	
2026.09.10	Martinez Alcantara, Jose Eduardo(265	ags	Sokolov, Ivan(2559)	ags	0-1	49	d20	
2026.08.23	Kris LeDoux	ags	Zion	ags	0-1	32	d20	
2026.08.23	David	ags	Kris LeDoux	ags	0-1	33	d20	
2026.07.19	Kris LeDoux	ags	Trey	ags	1-0	53	d20	
2025.06.02	Lei, Tingjie(2552)	ags	Vaishali, Rameshbabu(2475)	ags	1-0	46	d20	
2024.04.14	Rousseau	ags		ags	-	11	d20	
2017.04.27	Kolton	ags	Kris LeDoux	ags	1-0	23	d20	
2017.02.16	Matthew	ags	Kris LeDoux	ags	1-0	29	d20	

It will be empty if no pgn games have been analyzed, otherwise it will show all analyzed files in the "saved" folder. If analyzed games are present click the game and click analyze selected to open analysis window.

Click the Browse Files button to open either pgn or ags files from anywhere on the system. PGN files are chess game files and ags files are saved analysis of pgns. These ags files are created by analyzing pgn and are default saved in the "saved" directory.

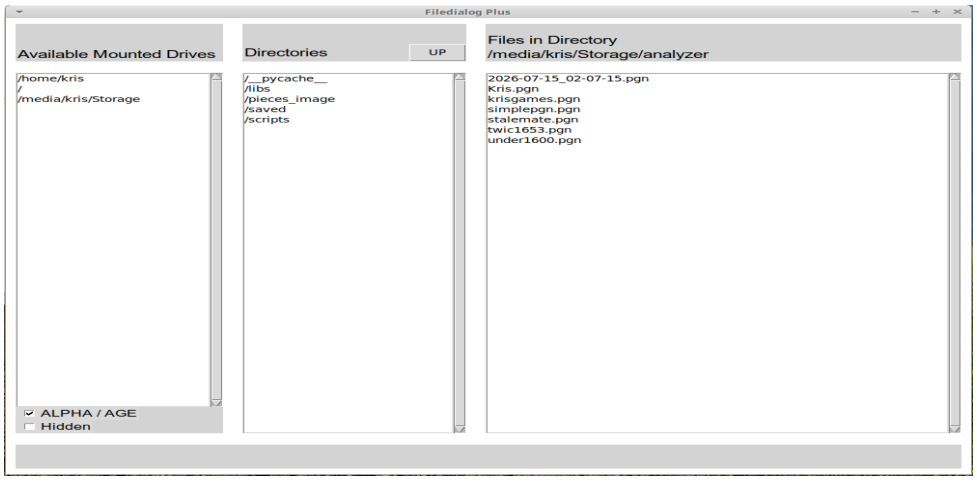
Here you can also select the depth to analyze when analyzing a PGN file. When viewing PGN files the AGS button will return you to the AGS saved files screen.

Either AGS or PGN list can be filtered to only display or Not display based on the filter setting. The filter can be just a few letters or whole names as desired to filter to what you want.

The Board button will open the interactive chessboard that can be used for many things.

- Entering a PGN move by move to be saved as pgn.
- Using Stockfish to graphically see the "best" three moves calculated in real time.
- Experiment with opening sequences using Stockfish guidance.

The Browse Files opens the Filedialogplus allowing you to search for pgn or ags files on your computer.



Search your system and find the file you want to open. If you open a pgn file it will be parsed for games and the games will be displayed in the pgn games viewer. If you open an ags file, the one game it contains will be displayed in the analysis window. We will go over pgn file next and this will lead to the analysis window.

If you open a pgn file it will be parsed for games and the games will be displayed in the pgn games viewer. This will show you an unfiltered view of the games in the file. Games in dark blue background have not been analyzed, games with light background are already in saved directory (analyzed).

You can filter by the names using the filter field. Enter your name or partial name in the filter box and then click Filter button and only those names will be displayed. Empty filter field and click Filter button to get them all again.

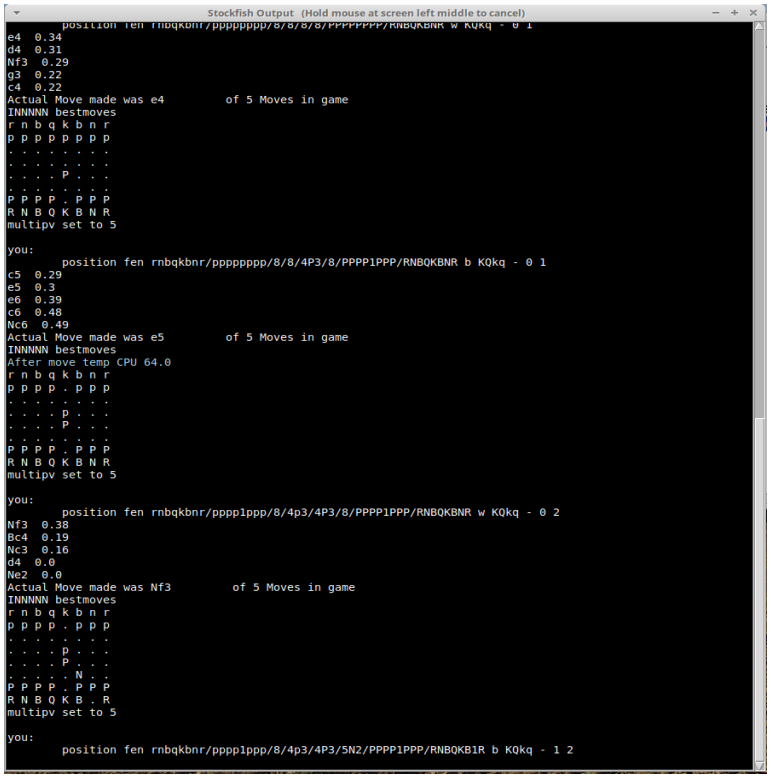
The Not checkbox allows you to filter by NOT displaying the names or partial names in the filter box but all others. You must always click the filter button to update the display. Case is important in the filter, so if you are not sure of case you can cheat and leave the first letter off.

The Depth to analyze pull down selects analysis depth. At the end of this manual I will go into depth (bore you) about CPU's, heat, and depth. For now, higher depth means better analysis but more time to analyze. This time can be a while. Lower depth is less quality analysis but faster.

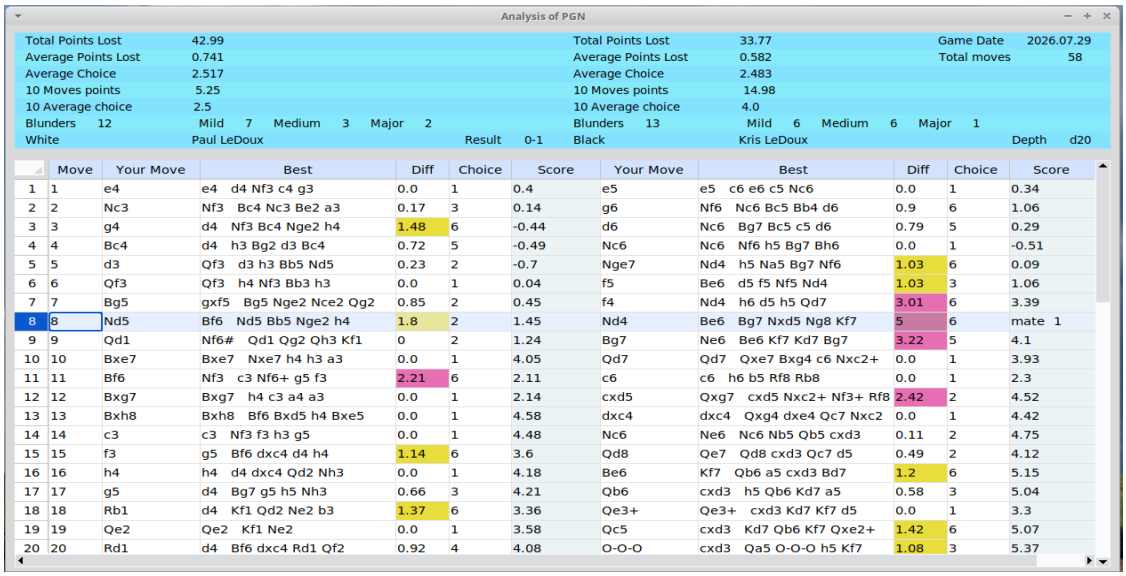
I have included a reasonable selection. This is subject to change. As of v1.0.0 selections are from 12 to 27 depth. This uses Stockfish 17.1 to analyze the game. The default multipv used is 5. This means that Stockfish will calculate the five best moves in best to worst order. Because of time to calculate, if you select a depth greater than 22 the multipv will be lowered to 2. This makes less Stockfish choices in the analysis, but speeds up the analysis some since 25 or 27 depth is CPU intensive and takes a while unless you have a super PC.

Select the game you want to analyze, and click Analyze Selected button. This will analyze the pgn, but only the main line and will ignore secondary lines.
This will bring up the Stockfish output window that shows the progress and minimizes when complete.
This may take from 1 to 25 minutes or more.

Once analysis has started, to stop (abort) analysis move mouse to left side of screen, middle top to bottom. This will tell Stockfish 17.1 to stop processing. You will have to keep the mouse there until the next move is calculated and Stockfish returns control to script. (1 to 30 seconds).



Once analyzed the analysis will be saved as an ags file to the "saved" directory. Once saved the analysis window will open. This is the same window that opens if you open an ags file instead of a pgn.



	Move	Your Move	Best	Diff	Choice	Score	Your Move	Best	Diff	Choice	Score
1	1	e4	e4 d4 Nf3 c4 g3	0.0	1	0.4	e5	e5 c6 e6 c5 Nc6	0.0	1	0.34
2	2	Nc3	Nf3 Bc4 Nc3 Be2 a3	0.17	3	0.14	g6	Nf6 Nc6 Bc5 Bb4 d6	0.9	6	1.06
3	3	g4	d4 Nf3 Bc4 Nge2 h4	1.48	6	-0.44	d6	Nc6 Bg7 Bc5 c5 d6	0.79	5	0.29
4	4	Bc4	d4 h3 Bg2 d3 Bc4	0.72	5	-0.49	Nc6	Nc6 Nf6 h5 Bg7 Bh6	0.0	1	-0.51
5	5	d3	Qf3 d3 h3 Bb5 Nd5	0.23	2	-0.7	Nge7	Nd4 h5 Na5 Bg7 Nf6	1.03	6	0.09
6	6	Qf3	Qf3 h4 Nf3 Bb3 h3	0.0	1	0.04	f5	Be6 d5 f5 Nf5 Nd4	1.03	3	1.06
7	7	Bg5	gxf5 Bg5 Nge2 Nce2 Qg2	0.85	2	0.45	f4	Nd4 h6 d5 h5 Qd7	3.01	6	3.39
8	8	Nd5	Bf6 Nd5 Bb5 Nge2 h4	1.8	2	1.45	Nd4	Be6 Bg7 Nxd5 Ng8 Kf7	5	6	mate 1
9	9	Qd1	Nf6# Qd1 Qg2 Qh3 Kf1	0	2	1.24	Bg7	Ne6 Be6 Kf7 Kd7 Bg7	3.22	5	4.1
10	10	Bxe7	Bxe7 Nxe7 h4 h3 a3	0.0	1	4.05	Qd7	Qd7 Qxe7 Bxg4 c6 Nxc2+	0.0	1	3.93
11	11	Bf6	Nf3 c3 Nf6+ g5 f3	2.21	6	2.11	c6	c6 h6 b5 Rf8 Rb8	0.0	1	2.3
12	12	Bxg7	Bxg7 h4 c3 a4 a3	0.0	1	2.14	cxd5	Qxg7 cxd5 Nxc2+ Nf3+ Rf8	2.42	2	4.52
13	13	Bxh8	Bxh8 Bf6 Bxd5 h4 Bxe5	0.0	1	4.58	dxc4	dxc4 Qxg4 dxe4 Qc7 Nxc2	0.0	1	4.42
14	14	c3	c3 Nf3 f3 h3 g5	0.0	1	4.48	Nc6	Ne6 Nc6 Nb5 Qb5 cxd3	0.11	2	4.75
15	15	f3	g5 Bf6 dxc4 d4 h4	1.14	6	3.6	Qd8	Qe7 Qd8 cxd3 Qc7 d5	0.49	2	4.12
16	16	h4	h4 d4 dxc4 Qd2 Nh3	0.0	1	4.18	Be6	Kf7 Qb6 a5 cxd3 Bd7	1.2	6	5.15
17	17	g5	d4 Bg7 g5 h5 Nh3	0.66	3	4.21	Qb6	cxd3 h5 Qb6 Kd7 a5	0.58	3	5.04
18	18	Rb1	d4 Kf1 Qd2 Ne2 b3	1.37	6	3.36	Qe3+	Qe3+ cxd3 Kd7 Kf7 d5	0.0	1	3.3
19	19	Qe2	Qe2 Kf1 Ne2	0.0	1	3.58	Qc5	cxd3 Kd7 Qb6 Kf7 Qxe2+	1.42	6	5.07
20	20	Rd1	d4 Bf6 dxc4 Rd1 Qf2	0.92	4	4.08	O-O-O	cxd3 Qa5 O-O-O h5 Kf7	1.08	3	5.37

This screen has two parts, the top which is statistics and game information, and the bottom which displays the moves of the analyzed game.

The top has game information and,

- Total points lost. Total of all moves difference between your choice and Stockfish best.
- Average points lost. Average by number of moves of above
- Average choice. Average of how far from Stockfish choice, 1 is same 2,3,4 5. 6 is not in choices. Lower is better, 1.0 would be identical to Stockfish. Higher number is worse.
- 10 Move points. Points lost (compared to Stockfish) in first ten moves.
- 10 average choice Like average choice but only for first ten moves of game.
- Blunders. Mild > .8 lost Medium > 1.8 lost major > 3.9 lost. Yellow pink red.

The bottom contains,

- Move number
- Your move. The move you made.
- Best. Stockfish moves listed from best to worst. 5 or 2 if depth > 22.
- Diff Points different from Stockfish best.
- Choice Ranking of your choice compared to Stockfish. 6 is out of multipv.
- Score Stockfish calculated score after this move.

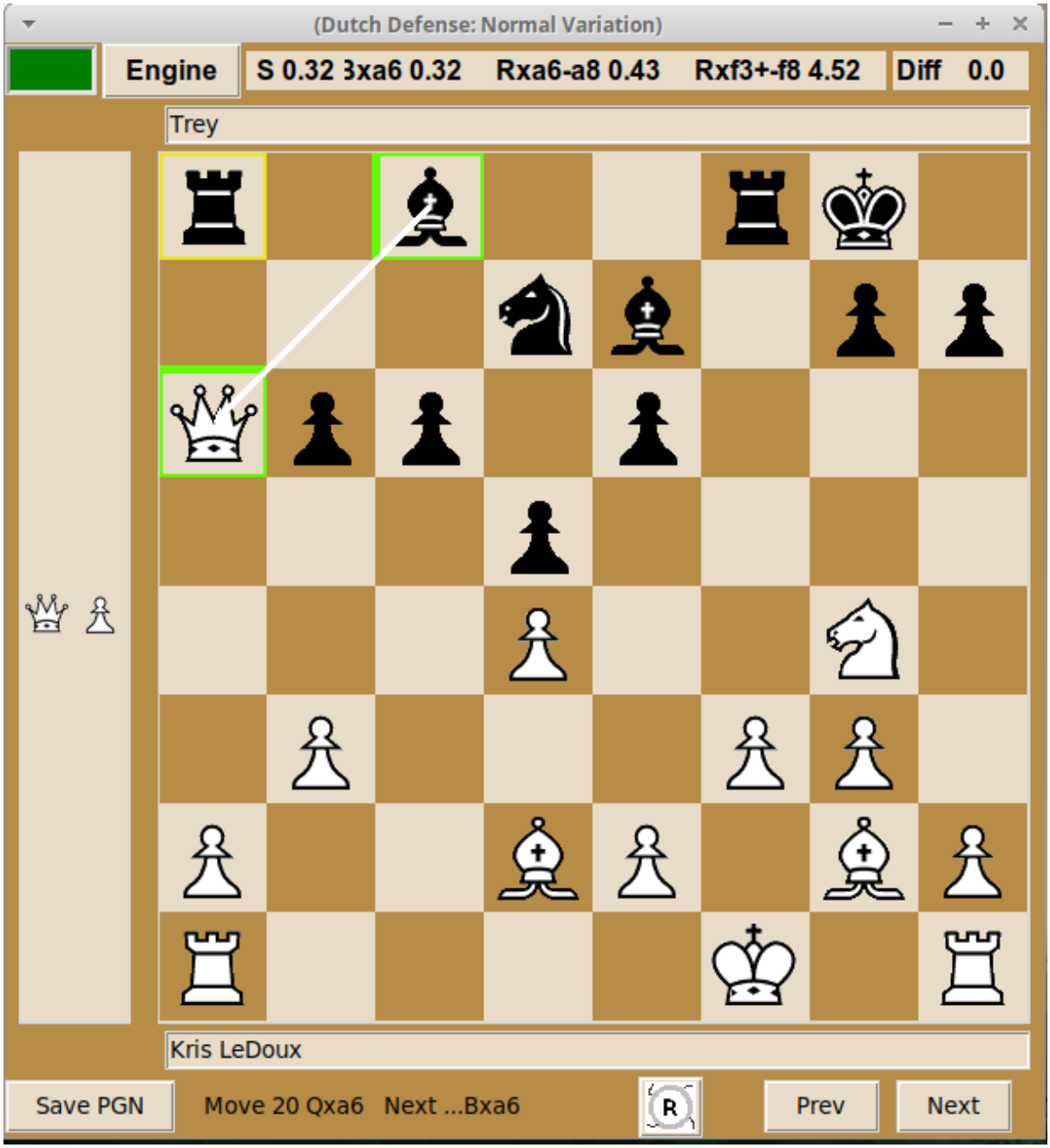
Blunders are marked in the list in colors. Yellow for minor blunders between .8 and 1.8. Pink for medium blunders between 1.81 and 3.9. Red for blunders greater than 3.9. Missed mates are marked as 5 points since I didn't know what value to give them.

You can scroll, view and to see the continuation click the far left column at the move number you want to see the continuation from.

Note on statistics.

I have found from a few comparisons, Grand masters have less blunders and are closer to choice of 1.8 to 2. You can kind of get a feel for a players rating from blunders and difference numbers. Lower rated players blunder and difference numbers are higher. Higher rated players have less blunders and almost no major blunders and their difference numbers are lower. These are not in depth statistics of course but interesting.

The continuation window.



The title of this window shows continuation opening. In the window the first box that is green when continuation window is opened is the "Assist" control. Red when off and green when assist is on. When on, any new move will trigger a calculation by Stockfish. The Engine button will only trigger calculation when pressed.

The display of scores is first S, the score after the move to be made. The next three scores are the Stockfish recommendations for the best three moves and their scores if you were to make them. These are graphically shown in green yellow and red for best to less best. If the difference from the fist to second or third moves is greater than .7 only the better ones are shown graphically. In above picture rxf3 is not graphically shown as it is a much lower scoring move.

The difference score at the top right is the difference OF THE NEXT MOVE between actual move made and recommended best move by Stockfish.

Buttons are New Prev and Next. Prev and Next are also linked to the right and left arrows on the keyboard. Bottom display shows current FEN move number, last move and next move if there is one.

Pressing esc key will close continuation window. Pressing b key closes continuation window and reloads it from the move it was begun with. This allows you to start over after calculating new continuations. The r key will rotate the display to change player perspective, same as the R button below. The Save PGN does just that.

You can also move a piece (who has the move). If you move it to the next move location it is just like the next button, if you move it to a new legal position it triggers a New Stockfish calculation based on that move. If you move a piece with the mouse and it goes back to it's original position the move is (I hope) illegal.

When calculating a "New" continuation Stockfish is told to calculate to a depth of 25 with multipv set to three. In complex positions this may slow appreciably when depth gets above 22. To stop the calculation at the next depth, move the mouse to the far left of the screen in the middle area top to bottom. Hold it there. When Stockfish gets to the next depth it will read this and stop calculating. Since Stockfish is not being run in a separate thread by the script, the script will not note the mouse location until Stockfish goes to it's next iteration (depth). Pyautogui will catch the mouse location when Stockfish comes up for air (returns control) and tell Stockfish to stop in an orderly fashion.

The Board (chessboard entry board)



This board is for experimentation and is similar to the continuation board. Numbers are slightly different since it does not have analyzed scores of continuation so it only shows score for the next move with no difference score.

It's default assist is off so that moves can be entered without triggering calculation. You can selectively calculate using the Engine button for one move or click the assist to turn it on.

This is convenient for practicing openings and seeing variations to get good moves and see if one you would make would be a blunder. You could enter first ten moves without assist and then go back and calculate them or any other way to experiment.

You can also enter a PGN and save it to a pgn file.

Extra crap the author want to say. Some of might be interesting.

Note on chess engines

Stockfish, like ANY chess engine is CPU intensive. CPU intensive use creates HEAT that must be dissipated in some way. If heat is not dissipated then CPU must be throttled (slowed) to keep from damaging itself. The operating system does this. For this reason my desktop PC with a slower CPU (but heat sink and fans) than my laptop analyses the same or faster than my laptop. Laptop's (not gaming laptop) are built for quiet and slimness so do not have large cpu heat sinks and high airflow fans that make more noise. While the CPU may be faster on paper sustained high CPU intensity tasks tend to slow down. Chess engines will generally show this. This will work fine on a laptop but may be slower than expected.

For systems that report CPU temperature (laptops typically do not) I have implemented some CPU protection. When the CPU temp is greater than 86 C a very short do nothing loop is triggered between each move calculation. If CPU temp goes above 93 C a several second do nothing kicks in after each move while CPU temp is greater than 93 C. For systems that do not report CPU temp (most laptops) the script will display 1 C and only the short delay is activated, never the longer delay as operating system must protect CPU.

There debate over whether CPU (chess engines) intensity can damage CPU's with no one truly knowing the definitive answer. My inclusion of this only slows systems appreciably if temp over 93 C. On a well cooled system this won't happen often and on a laptop that does not report temperature it will never happen.

I have chosen to "protect" (might not be needed) the cpu by slowing down when temperature gets over 93. it doesn't add that much time to the analysis and I figure why not.

On the same note I set multipv to 2 when higher than 22 depth to save CPU heat. I may change this in the future as CPU's get faster. For now and with my CPU it works. I really like the multipv of five so may change that and you just get stuck with the time penalty.